

# Expt 4 Design of Low Pass Filter using Windowing Techniques

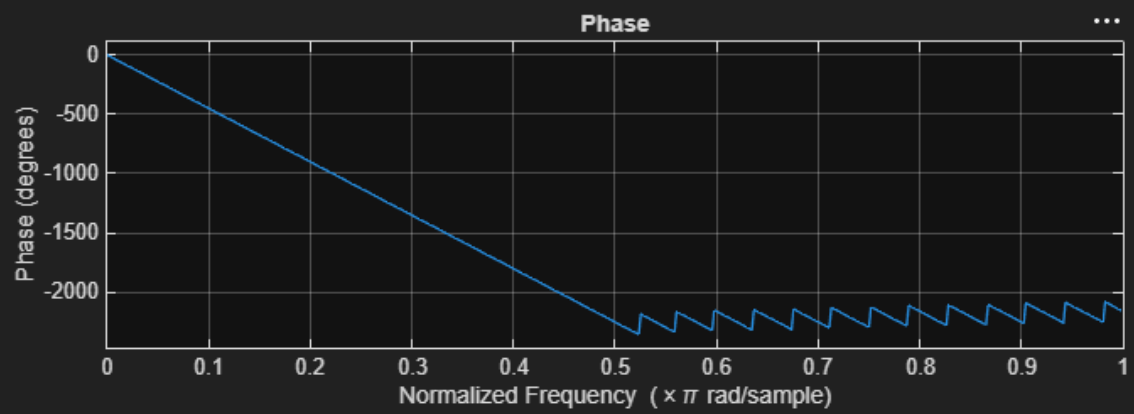
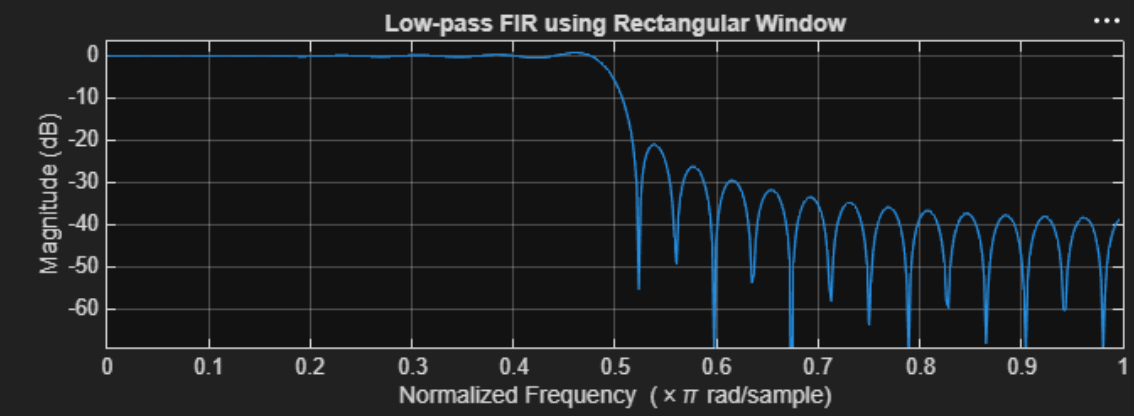
## 4.1 Design and Analysis of an FIR Low-Pass Filter Using the Rectangular Window Technique

```
1  clc;
2  clear;
3  close all;
4  %% --- Input Parameters ---
5  fp = 1000; % Passband cutoff (Hz)
6  fs = 4000; % Sampling frequency (Hz)
7  N = 50; % Filter order (even number preferred)
8
9  wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)
10 %% --- 1. FIR Low Pass using Rectangular Window ---
11 b_rect = fir1(N, wc, 'low', rectwin(N+1));
12 disp('--- Rectangular Window Coefficients ---');
13 disp(b_rect);
14 %% --- 2. FIR Low Pass using Hamming Window ---
15 b_hamm = fir1(N, wc, 'low', hamming(N+1));
16 disp('--- Hamming Window Coefficients ---');
17 disp(b_hamm);
18 %% --- 3. FIR Low Pass using Hanning Window ---
19 b_hann = fir1(N, wc, 'low', hanning(N+1));
20 disp('--- Hanning Window Coefficients ---');
21 disp(b_hann);
22 %% --- Plot Frequency Responses ---
23 figure;
24 freqz(b_rect,1);
25 title('Low-pass FIR using Rectangular Window');
26 figure;
27 freqz(b_hamm,1);
28 title('Low-pass FIR using Hamming Window');
29 figure;
30 freqz(b_hann,1);
31 title('Low-pass FIR using Hanning Window');
32 |
```

Command Window

New to MATLAB? See resources for [Getting Started](#).

```
--- Rectangular Window Coefficients ---
0.0126
0
-0.0137
0
0.0150
0
-0.0166
0
0.0185
0
-0.0210
0
0.0242
```



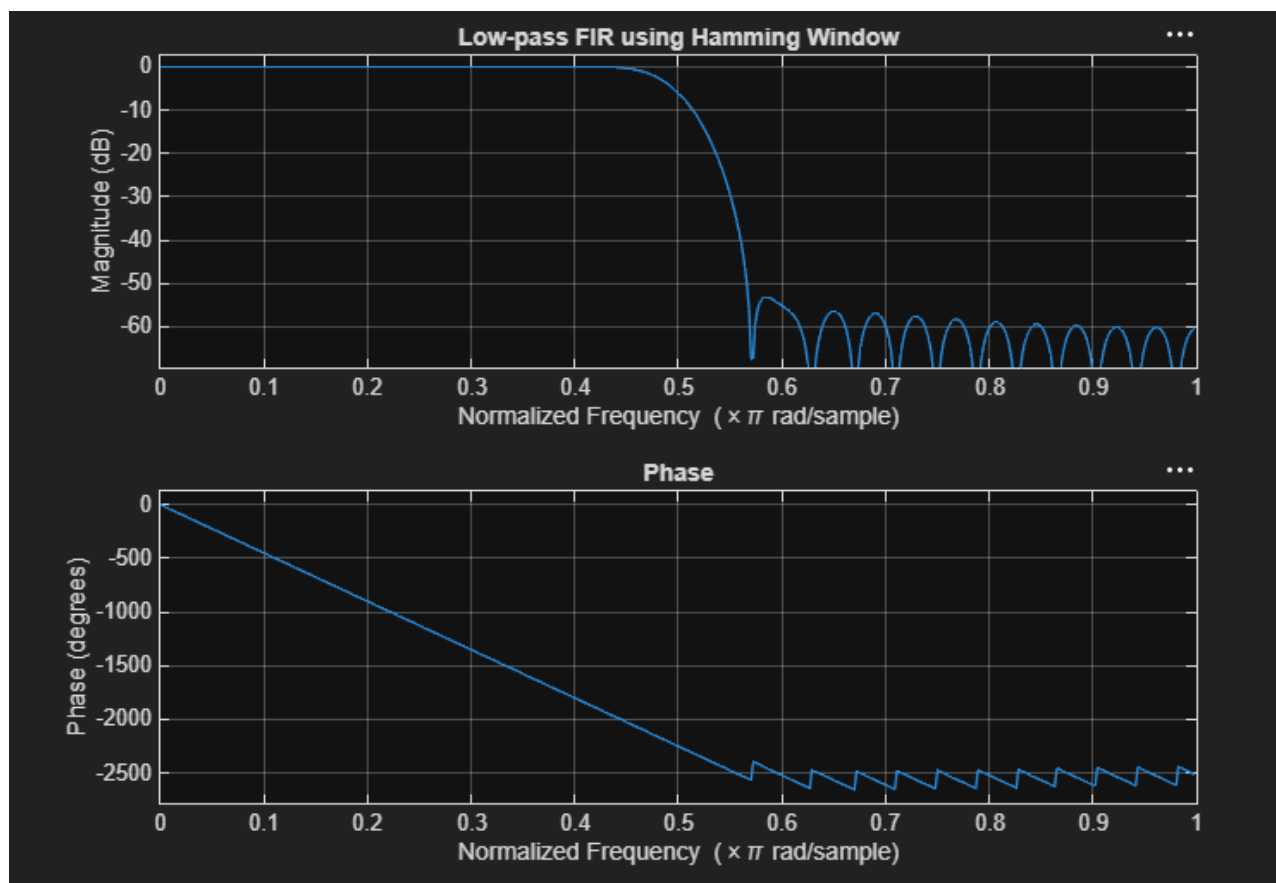
## 4.2 Design and Analysis of an FIR Low-Pass Filter Using the Hamming Window Technique

```
1  clc;
2  clear;
3  close all;
4  %% --- Input Parameters ---
5  fp = 1000; % Passband cutoff (Hz)
6  fs = 4000; % Sampling frequency (Hz)
7  N = 50; % Filter order (even number preferred)
8  wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)
9  %% --- FIR Low Pass using Hamming Window ---
10
11  b_hamm = fir1(N, wc, 'low', hamming(N+1));
12  disp('--- Hamming Window Coefficients ---');
13  disp(b_hamm);
14  %% --- Plot Frequency Responses ---
15  figure;
16  freqz(b_hamm,1);
17  title('Low-pass FIR using Hamming Window');
```

Command Window

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```
0
-0.0084
0
0.0055
0
-0.0034
0
0.0021
0
-0.0013
0
0.0010
```



### 4.3 Design and Analysis of an FIR Low-Pass Filter Using the Hanning Window Technique

```
1  clc;
2  clear;
3  close all;
4  %% --- Input Parameters ---
5  fp = 1000; % Passband cutoff (Hz)
6  fs = 4000; % Sampling frequency (Hz)
7  N = 50; % Filter order (even number preferred)
8  wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)
9
10 %% --- FIR Low Pass using Hanning Window ---
11 b_hann = fir1(N, wc, 'low', hanning(N+1));
12 disp('--- Hanning Window Coefficients ---');
13 disp(b_hann);
14 %% --- Plot Frequency Responses ---
15 figure;
16 freqz(b_hann,1);
17 title('Low-pass FIR using Hanning Window');
```

Command Window

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```
0
-0.0081
0
0.0050
0
-0.0028
0
0.0013
0
-0.0004
0
0.0000
>>
```

